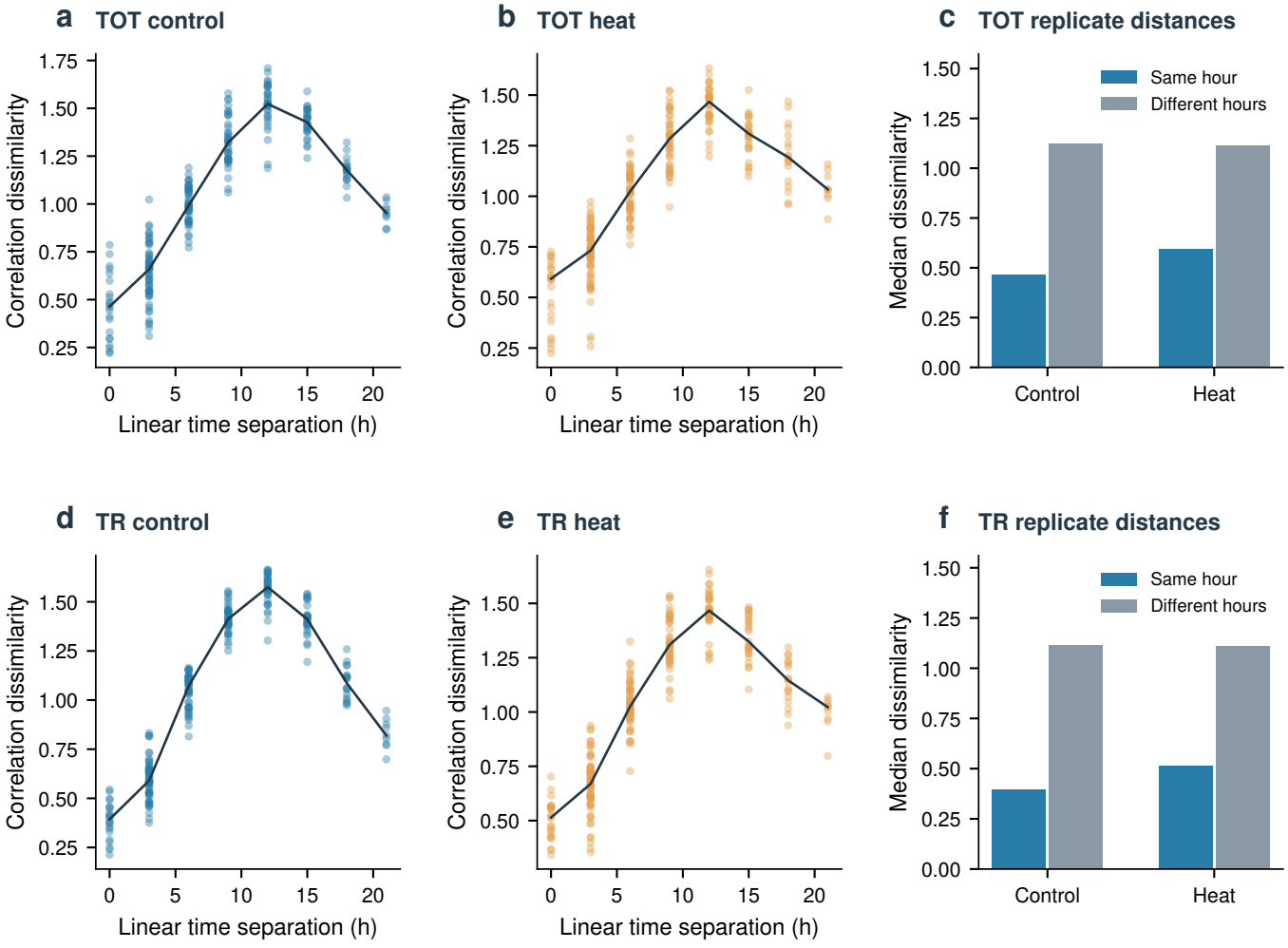
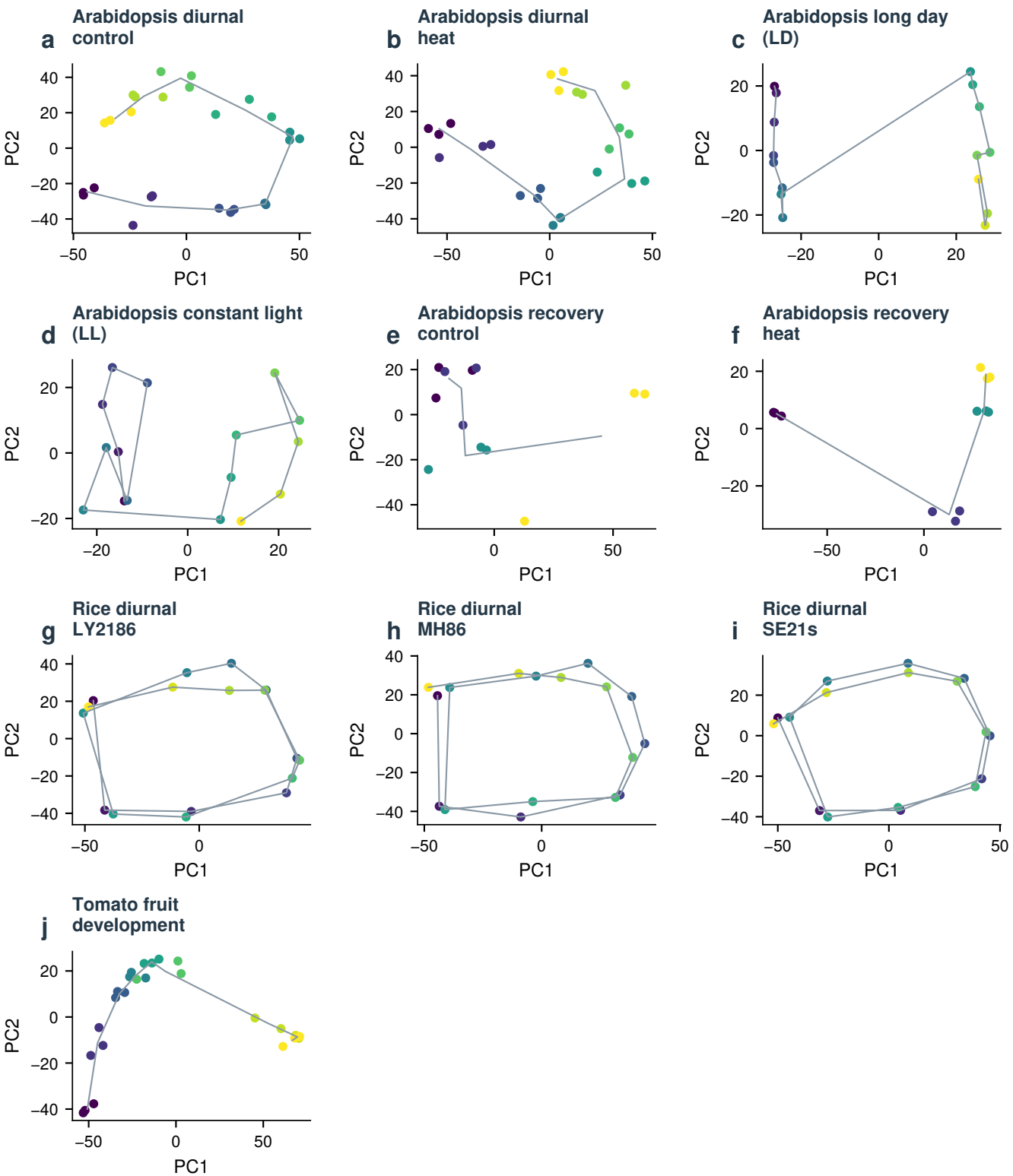
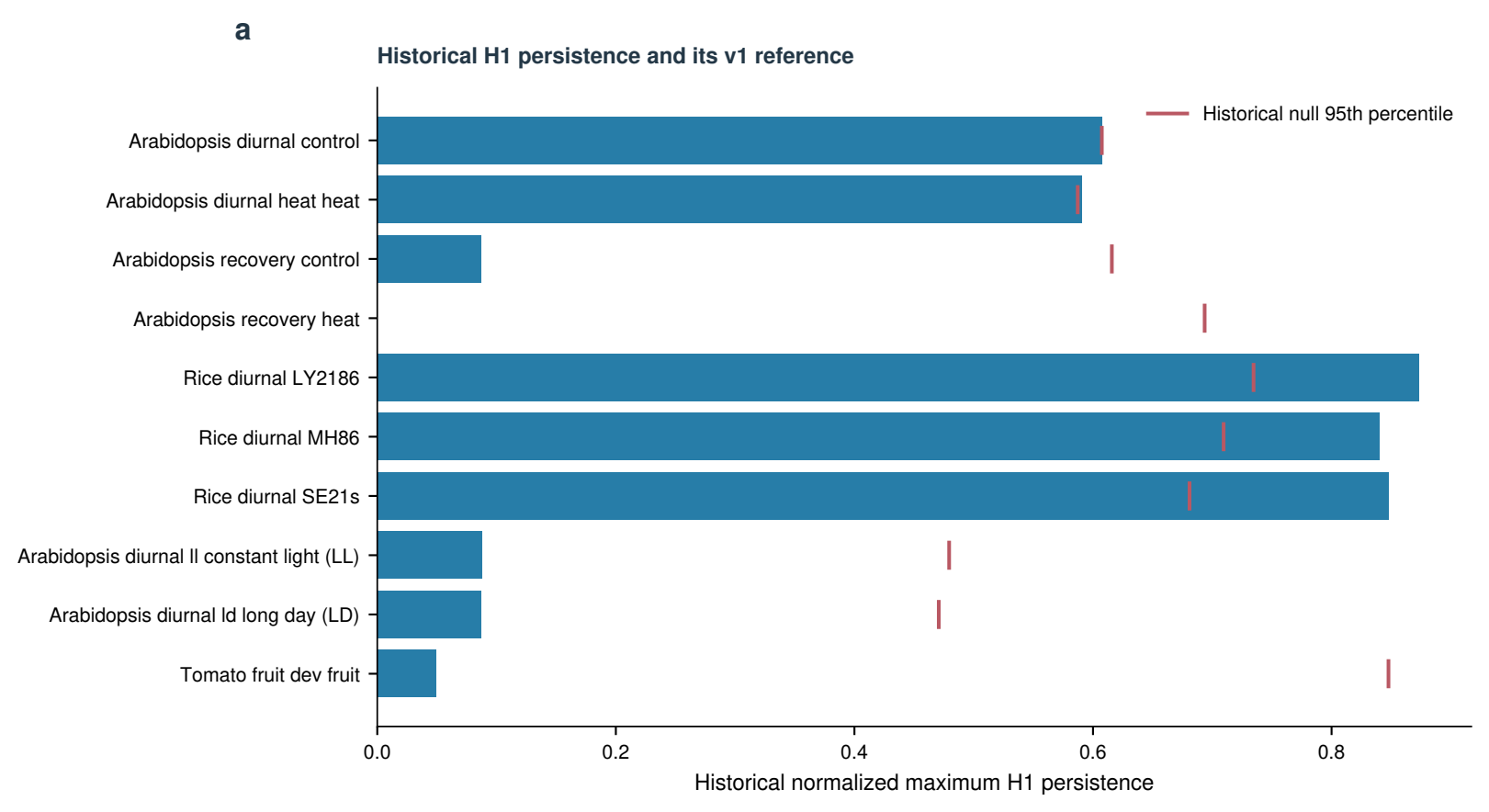
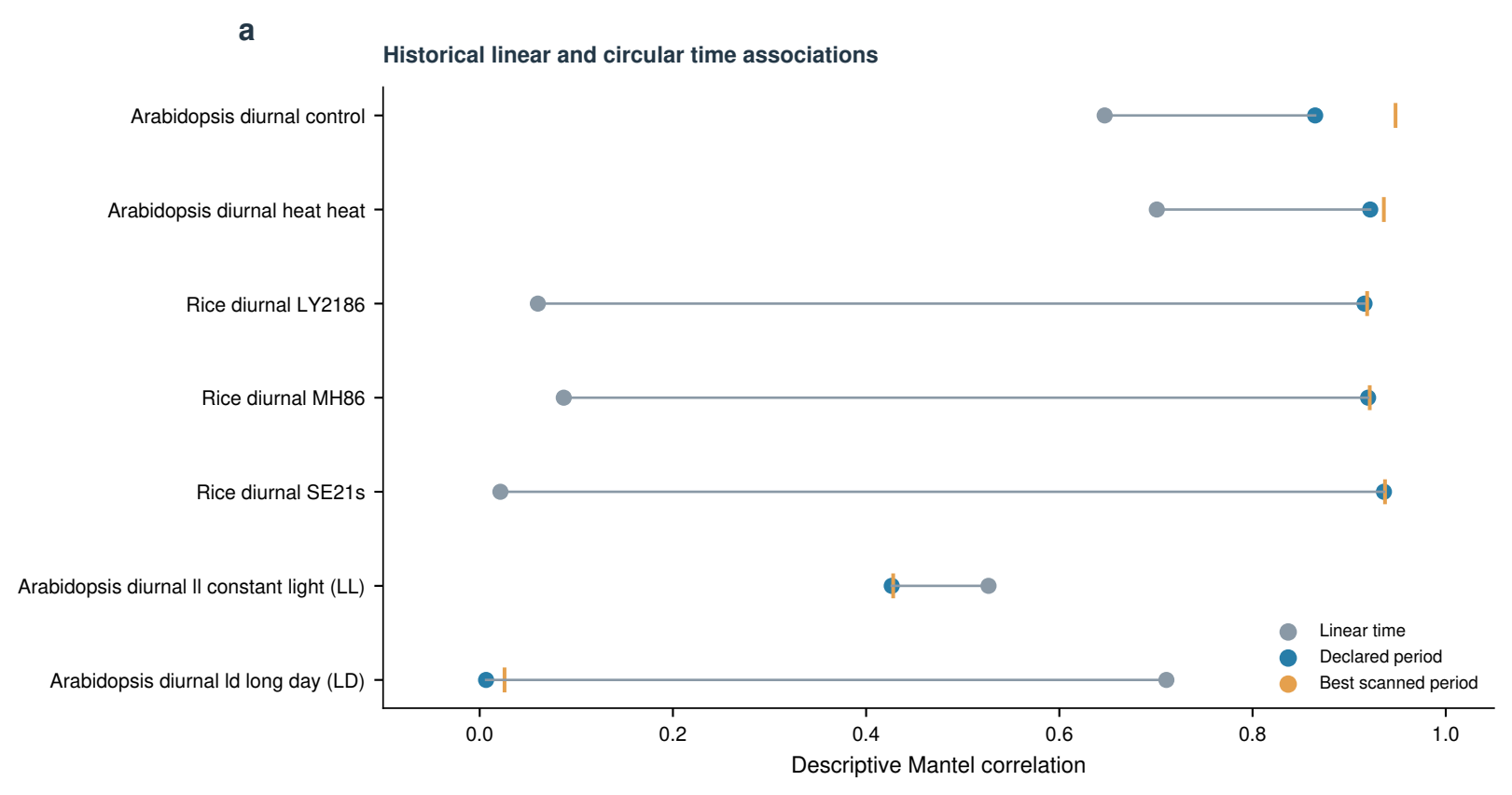
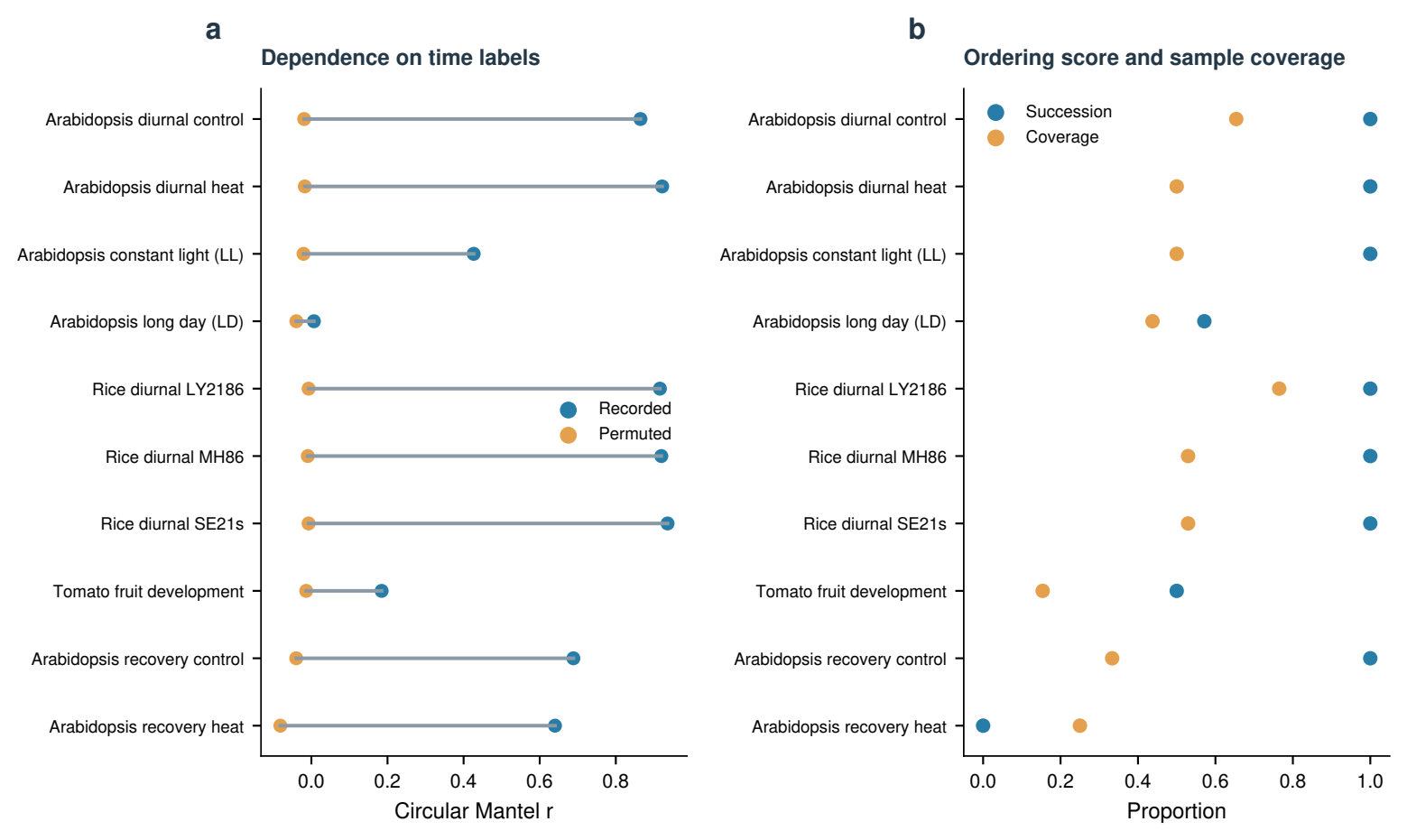
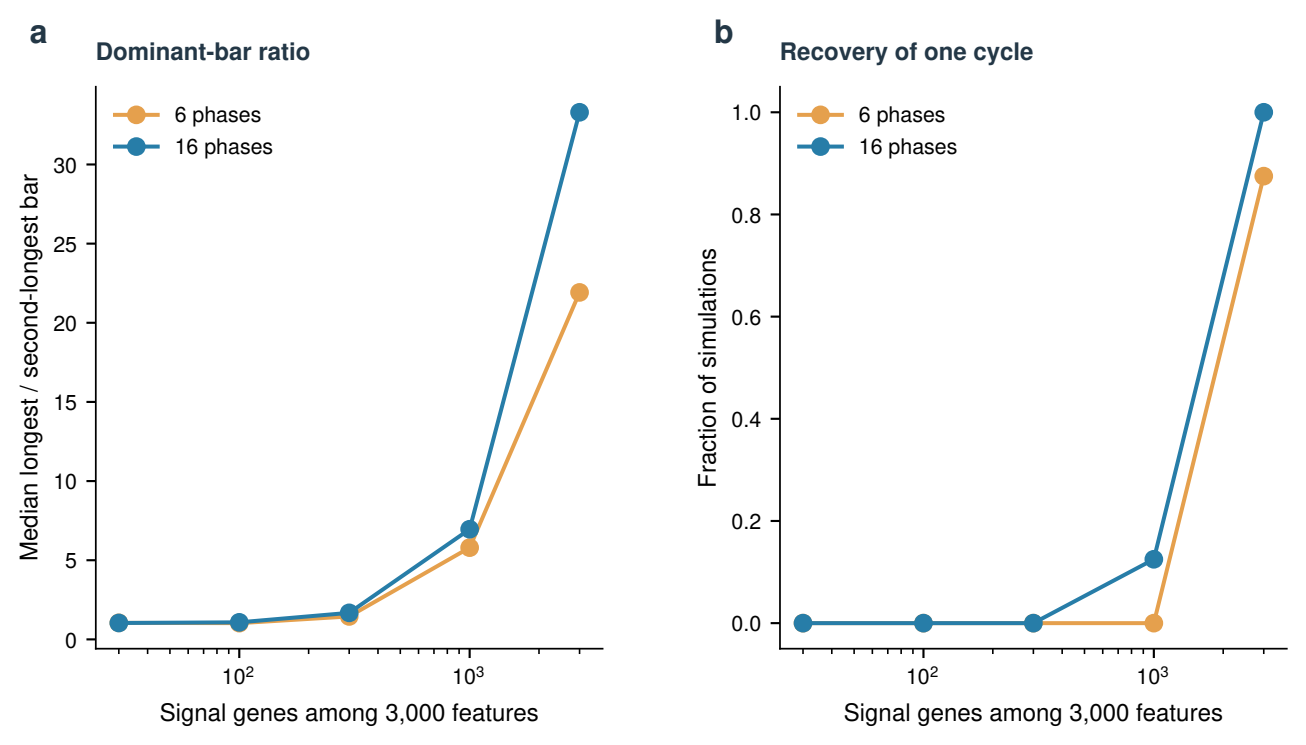

a
TOT control
Correlation dissimilarity
Linear time separation (h)
b
TOT heat
Correlation dissimilarity
Linear time separation (h)
c
TOT replicate distances
Median dissimilarity
Same hour
Different hours
Control
Heat
d
TR control
Correlation dissimilarity
Linear time separation (h)
e
TR heat
Correlation dissimilarity
Linear time separation (h)
f
TR replicate distances
Median dissimilarity
Same hour
Different hours
Control
Heat

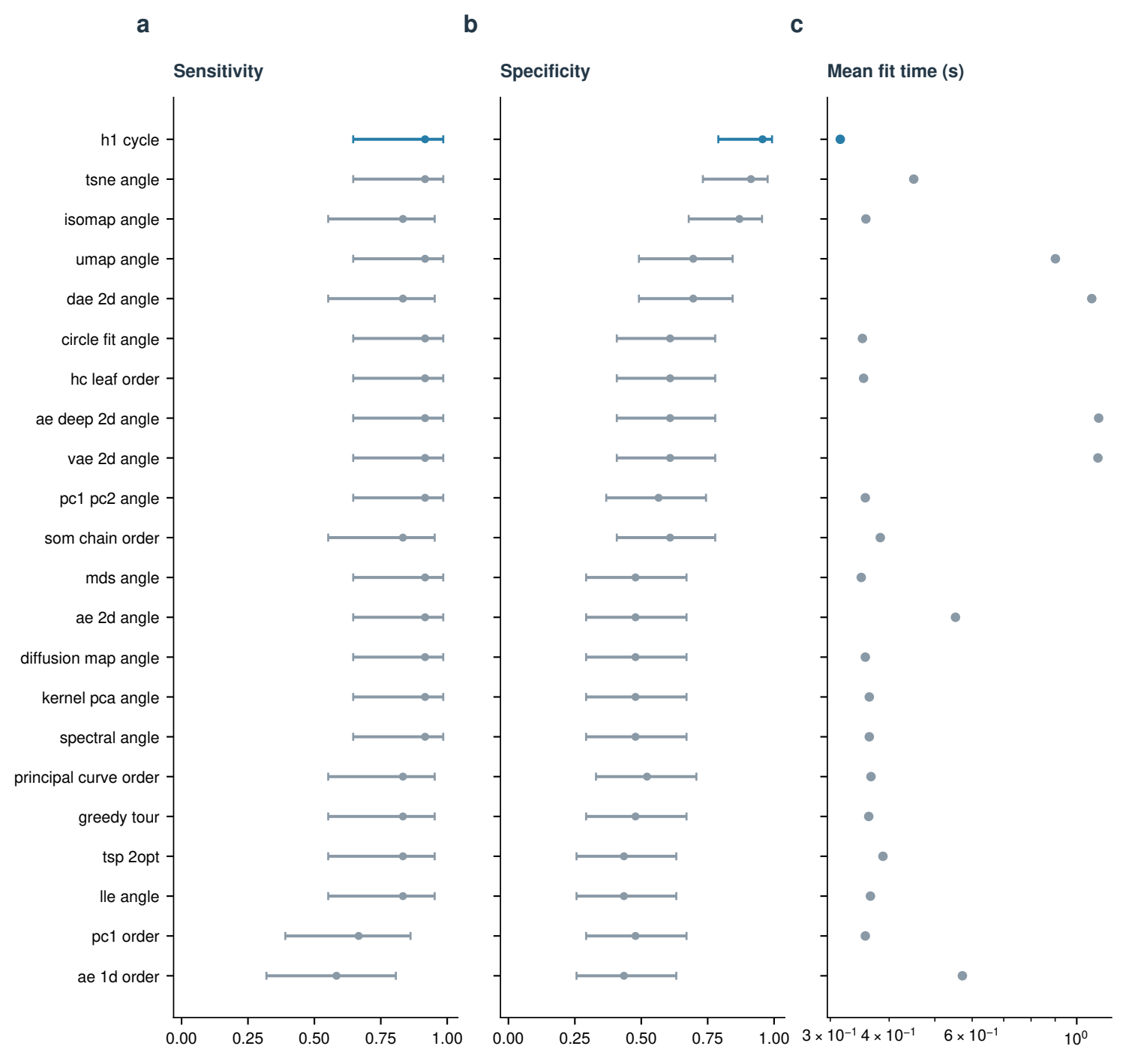

a
Arabidopsis diurnal control
b
Arabidopsis diurnal heat
c
Arabidopsis long day (LD)
d
Arabidopsis constant light (LL)
e
Arabidopsis recovery control
f
Arabidopsis recovery heat
g
Rice diurnal LY2186
h
Rice diurnal MH86
i
Rice diurnal SE21s
j
Tomato fruit development
PC1
PC2

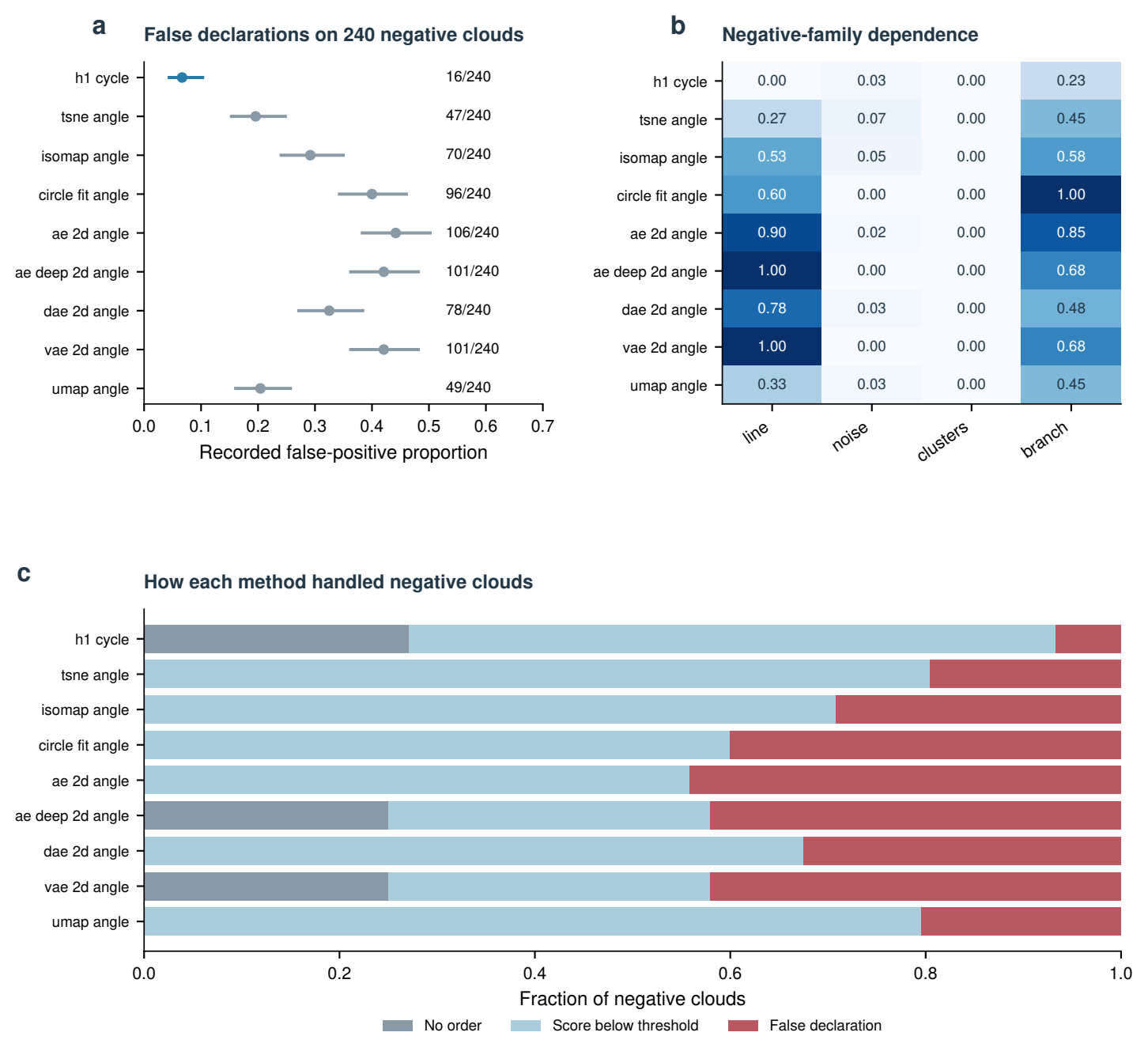

a
Historical H1 persistence and its v1 reference
Historical null 95th percentile
Arabidopsis diurnal control
Arabidopsis diurnal heat heat
Arabidopsis recovery control
Arabidopsis recovery heat
Rice diurnal LY2186
Rice diurnal MH86
Rice diurnal SE21s
Arabidopsis diurnal ll constant light (LL)
Arabidopsis diurnal ld long day (LD)
Tomato fruit dev fruit
0.0
0.2
0.4
0.6
0.8
Historical normalized maximum H1 persistence

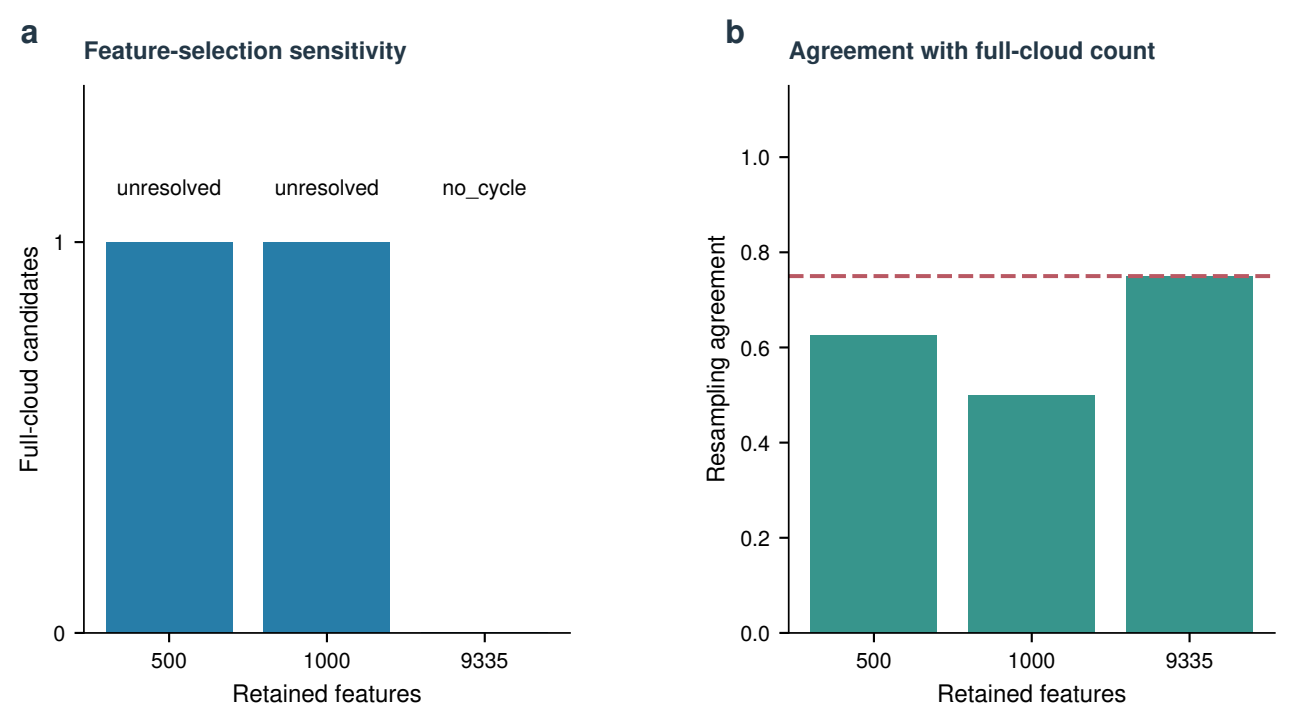

a
Historical linear and circular time associations
Arabidopsis diurnal control
Arabidopsis diurnal heat heat
Rice diurnal LY2186
Rice diurnal MH86
Rice diurnal SE21s
Arabidopsis diurnal ll constant light (LL)
Arabidopsis diurnal ld long day (LD)
Linear time
Declared period
Best scanned period
0.0
0.2
0.4
0.6
0.8
1.0
Descriptive Mantel correlation

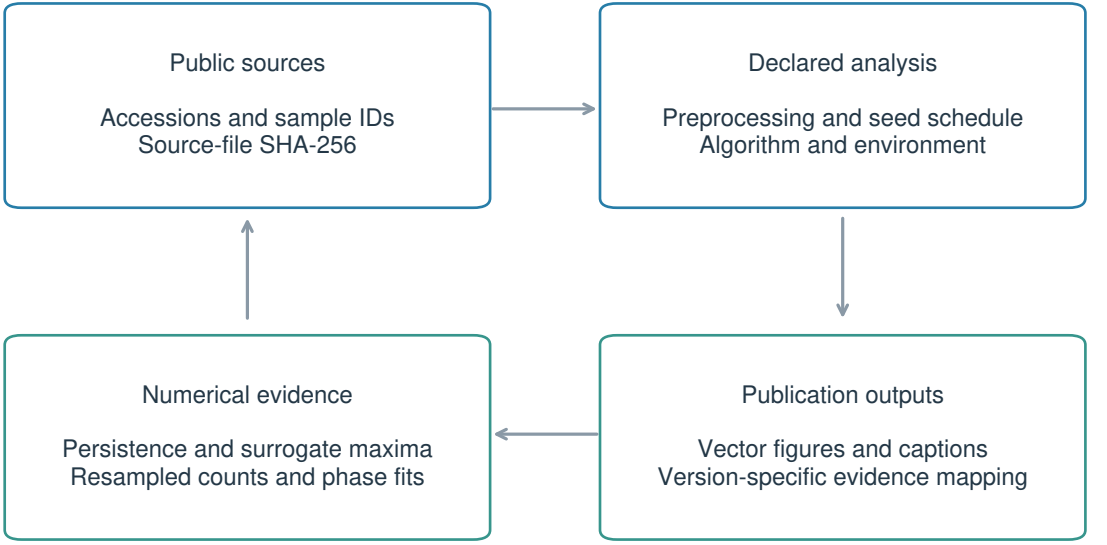

a
Dependence on time labels
Arabidopsis diurnal control
Arabidopsis diurnal heat
Arabidopsis constant light (LL)
Arabidopsis long day (LD)
Rice diurnal LY2186
Rice diurnal MH86
Rice diurnal SE21s
Tomato fruit development
Arabidopsis recovery control
Arabidopsis recovery heat
Recorded
Permuted
0.0
0.2
0.4
0.6
0.8
Circular Mantel r
b
Ordering score and sample coverage
Succession
Coverage
0.0
0.2
0.4
0.6
0.8
1.0
Proportion

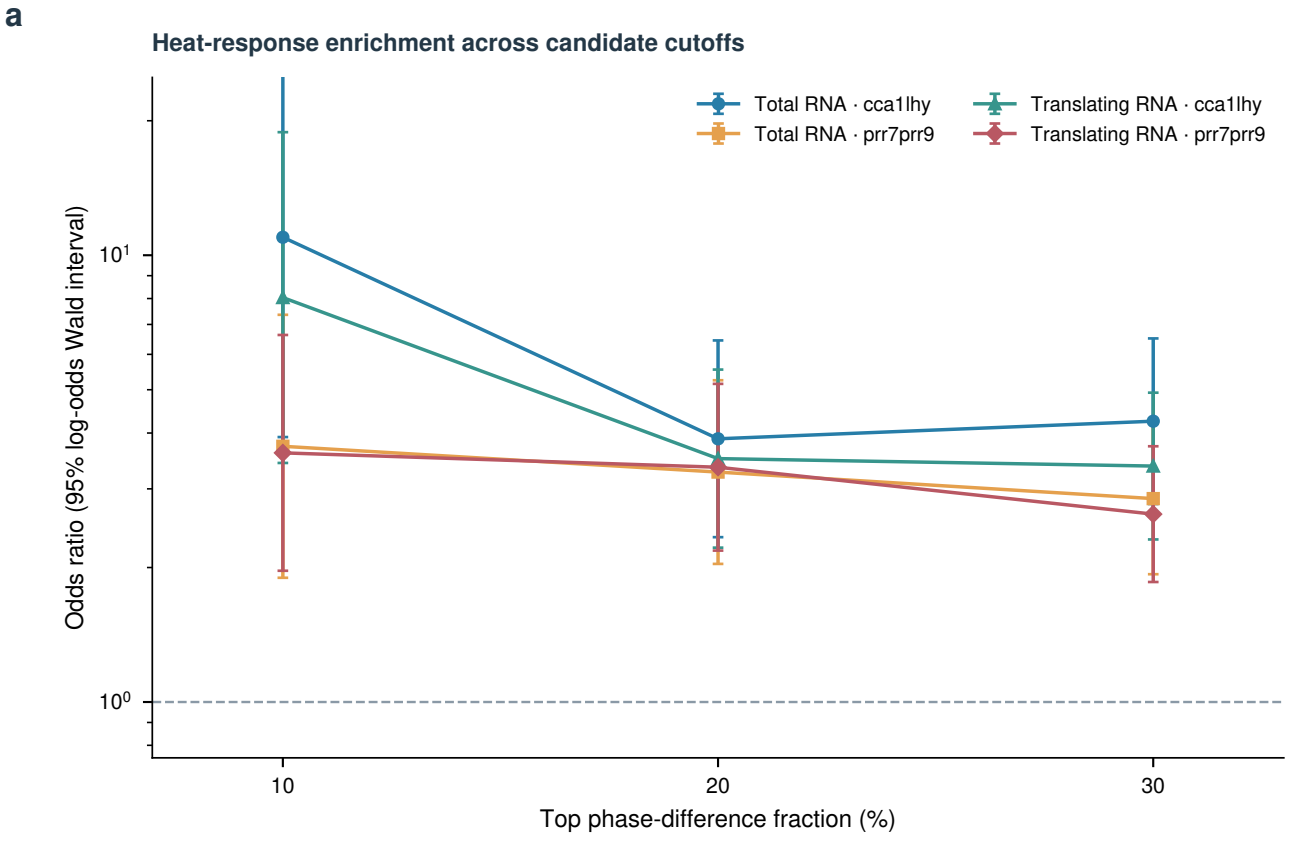

a
Dominant-bar ratio
6 phases
16 phases
Median longest / second-longest bar
0
5
10
15
20
25
30
10^2
10^3
Signal genes among 3,000 features
b
Recovery of one cycle
6 phases
16 phases
Fraction of simulations
0.0
0.2
0.4
0.6
0.8
1.0
10^2
10^3
Signal genes among 3,000 features


a
b
c
Sensitivity
Specificity
Mean fit time (s)
h1 cycle
tsne angle
isomap angle
umap angle
dae 2d angle
circle fit angle
hc leaf order
ae deep 2d angle
vae 2d angle
pc1 pc2 angle
som chain order
mds angle
ae 2d angle
diffusion map angle
kernel pca angle
spectral angle
principal curve order
greedy tour
tsp 2opt
lle angle
pc1 order
ae 1d order
0.00
0.25
0.50
0.75
1.00
0.00
0.25
0.50
0.75
1.00
3 × 10−1
4 × 10−1
6 × 10−1
10^0


a
False declarations on 240 negative clouds
h1 cycle
tsne angle
isomap angle
circle fit angle
ae 2d angle
ae deep 2d angle
dae 2d angle
vae 2d angle
umap angle
16/240
47/240
70/240
96/240
106/240
101/240
78/240
101/240
49/240
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Recorded false-positive proportion
b
Negative-family dependence
0.00 0.03 0.00 0.23
0.27 0.07 0.00 0.45
0.53 0.05 0.00 0.58
0.60 0.00 0.00 1.00
0.90 0.02 0.00 0.85
1.00 0.00 0.00 0.68
0.78 0.03 0.00 0.48
1.00 0.00 0.00 0.68
0.33 0.03 0.00 0.45
line
noise
clusters
branch
c
How each method handled negative clouds
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of negative clouds
No order
Score below threshold
False declaration


a
Feature-selection sensitivity
unresolved
unresolved
no_cycle
Full-cloud candidates
1
0
500
1000
9335
Retained features
b
Agreement with full-cloud count
Resampling agreement
1.0
0.8
0.6
0.4
0.2
0.0
500
1000
9335
Retained features


Public sources
Accessions and sample IDs
Source-file SHA-256
Declared analysis
Preprocessing and seed schedule
Algorithm and environment
Publication outputs
Vector figures and captions
Version-specific evidence mapping
Numerical evidence
Persistence and surrogate maxima
Resampled counts and phase fits


a
Heat-response enrichment across candidate cutoffs
Total RNA · cca1lhy
Total RNA · prr7prr9
Translating RNA · cca1lhy
Translating RNA · prr7prr9
Odds ratio (95% log-odds Wald interval)
10^1
10^0
10
20
30
Top phase-difference fraction (%)